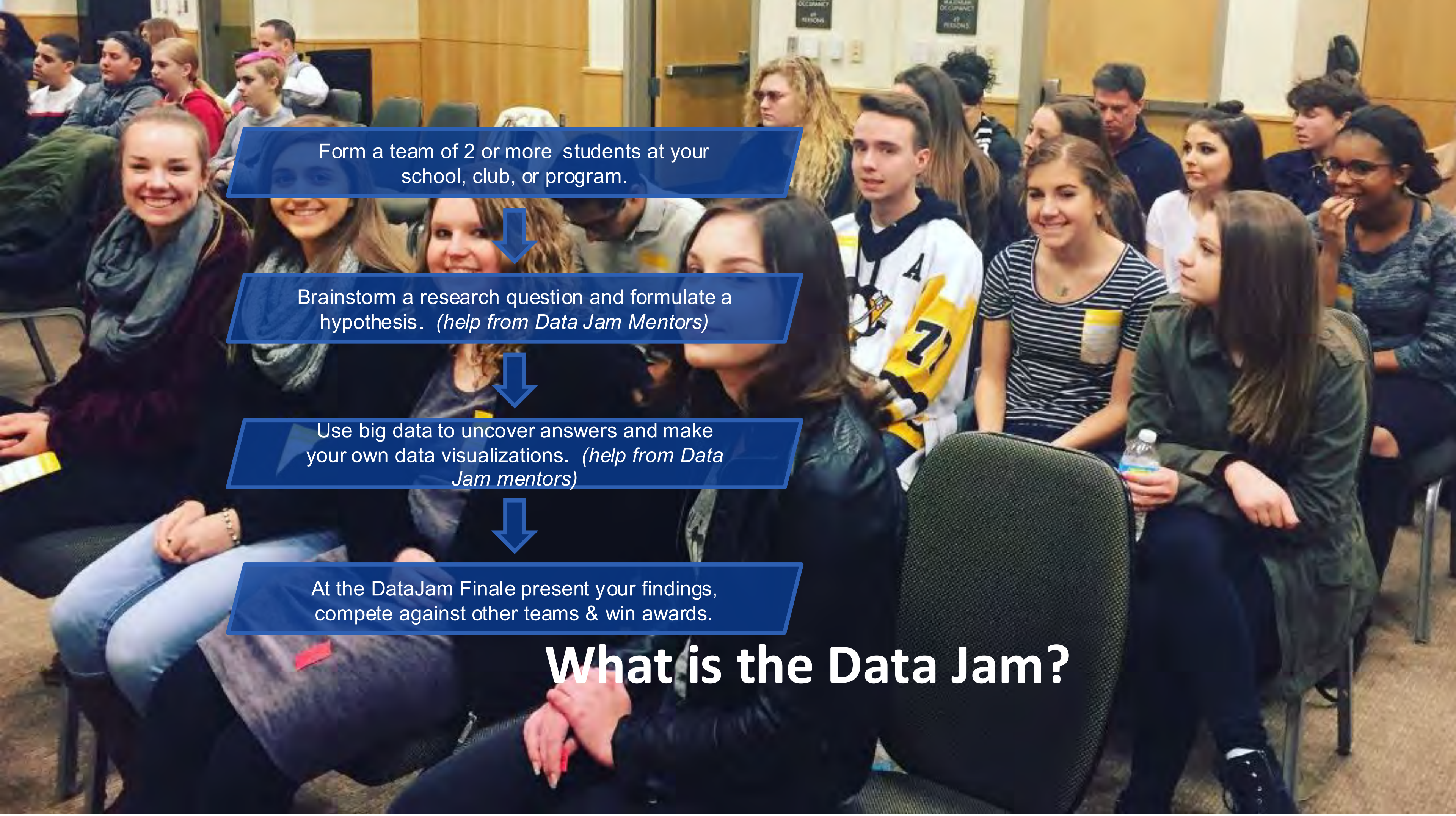
DataJam Advisor Workshop

THE DATAJAM



Form a team of 2 or more students at your school, club, or program.

Brainstorm a research question and formulate a hypothesis. *(help from Data Jam Mentors)*

Use big data to uncover answers and make your own data visualizations. *(help from Data Jam mentors)*

At the DataJam Finale present your findings, compete against other teams & win awards.

What is the Data Jam?

Why is *DataJam* so engaging?

Students choose a question they are interested in

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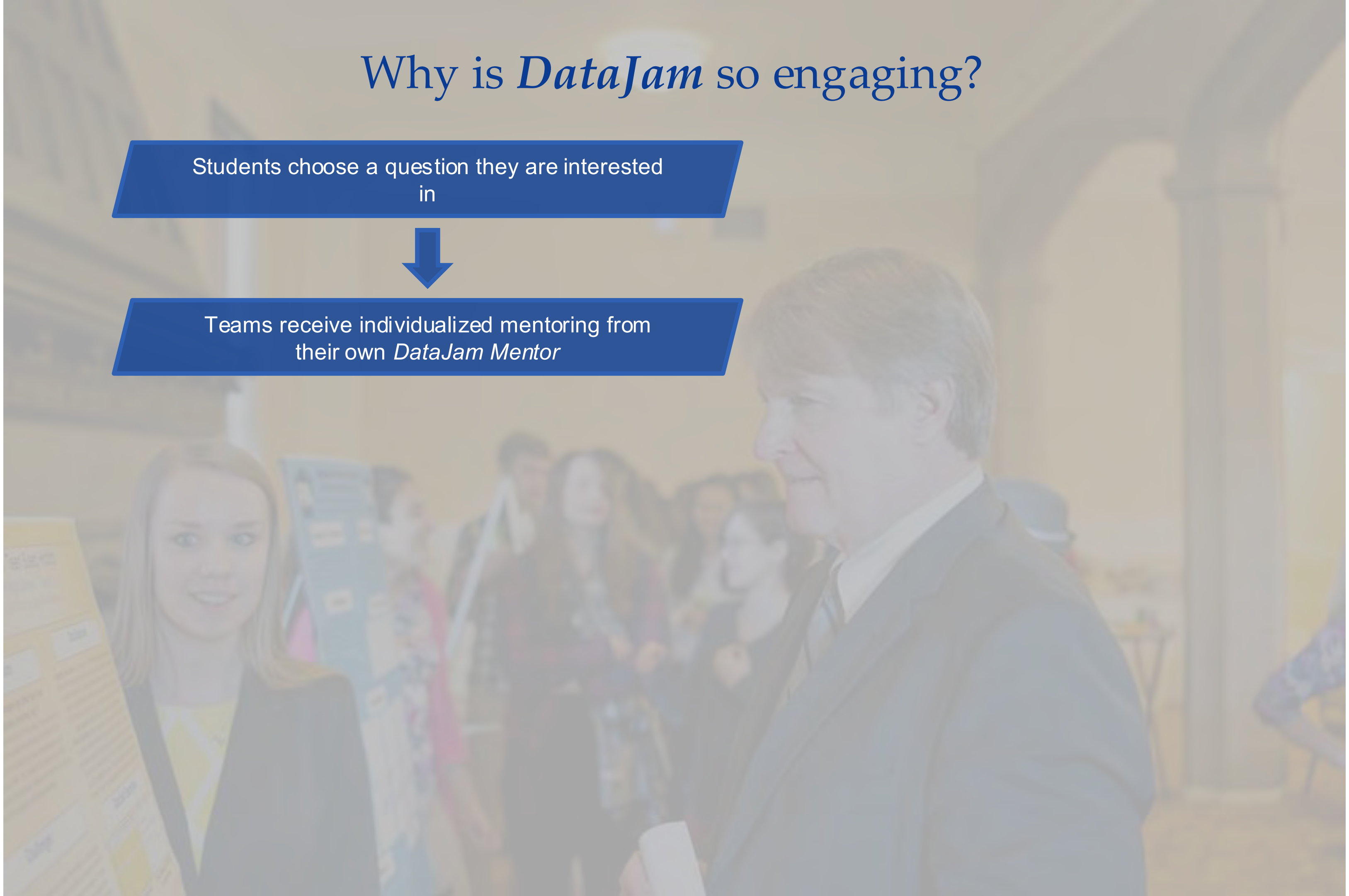
Teams receive individualized mentoring from their own *DataJam Mentor*



The DataJam In Brief

Excitement

A competition focusing on **the process and uses of big data** to **answer research questions**

Skills

Formulate a research question **use case**,, find **data, analyze** their data,, make data **visualizations**,, tell a **data story** and **present** findings to a panel of data science judges.. .

Resources

Free resources to High School students and afterschool programs,, such as the Boys & Girls Clubs,, **reduces barriers to Data Science career entry**..

The Mentors

Training

The DataJam Mentors are college students interested in the **power of data** and data **analysis** who are **trained** to work with **diverse** communities and needs..

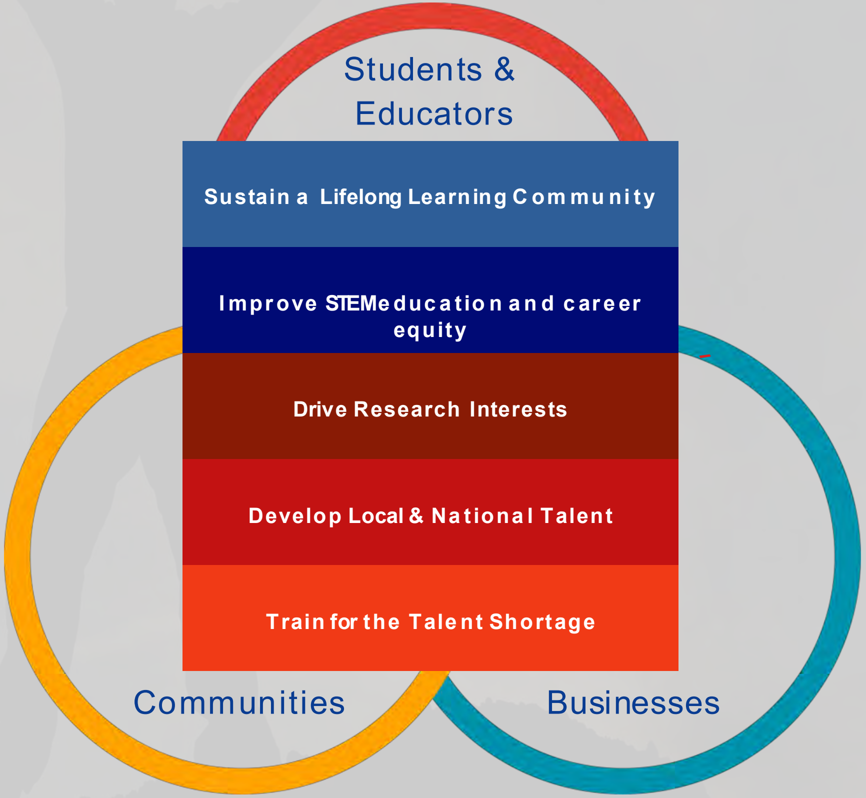
Culture

DataJam Mentors volunteer their time,, creating a lifetime journey of **community** and **inspiration** based on **trust** and new ways of working..

The DataJam Goals

By providing awareness, access and community to students, especially those with minimal resources,
we all win!

Win-Win-Win



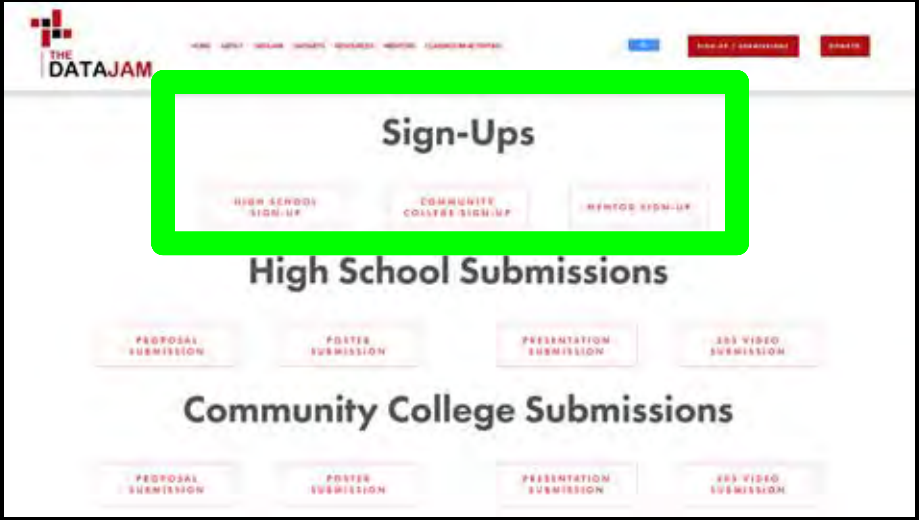
Find out more at **thedatajam.org**

How to Sign-Up your DataJam Team!

What you'll need:

- Student Names and eMails
- Team Number you assign to each team - even if there is only one
- Availability of Teams to meet with a Mentor

Let's do an example!



Adjusting Teams - Adding or Removing Students

Instead of filling out a new form, simply email us your changes! (datajam@thedatajam.org)

We understand the need for changes happens. If you think you will have many changes, please wait until you have several and let us know all at once.

Permission Slips – Filling them out and submitting them!

For high school students, permission slips must be filled out by each student and signed by a parent or guardian.

How permission slips are submitted is up to you! Permission slips can be submitted by each student or collected and combined and submitted by you. Completed permission slips are emailed to datajam@thedatajam.org.

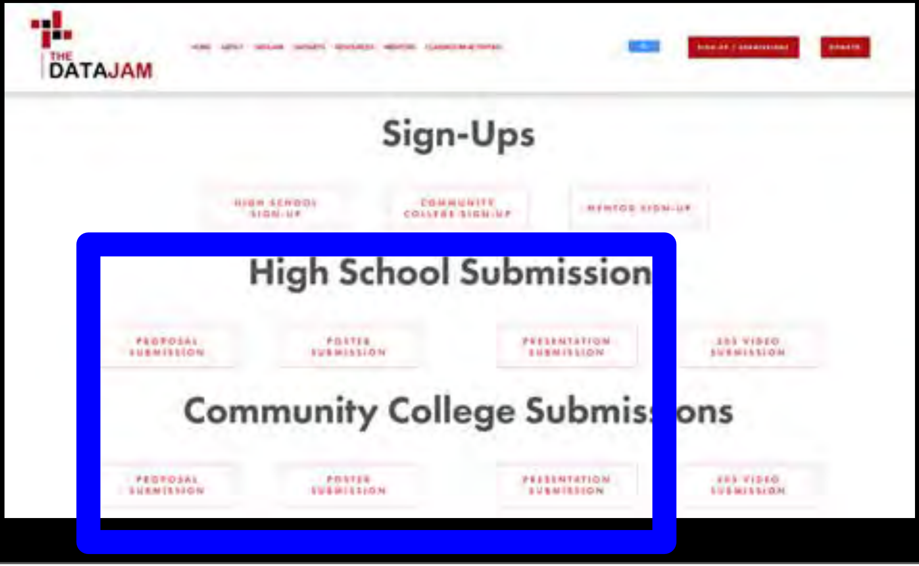
Let’s do one together!

Proposals/Posters/Presentations

Submitting documents for the DataJam just got easier!

Now, teams or advisors submit their documents online! Each document should be submitted as a pdf.

We request that documents are filled out with your organization name and the teams name or number so we can quickly identify it.



Optimizing Work with Your Team's DataJam Mentor!

- DataJam Mentors are university students who have received specialized training to provide mentorship for the DataJam.
- Mentors will meet with your students by videoconference and using The DataJam SLACK workspace.
- It works best to arrange a regular time for the team to meet with the mentor.
- The mentor will provide regular summaries of their work with your students on a timeframe (e.g., weekly) agreed on by you and the mentor.
- Mentors can help with *every step* of The DataJam process.

Facilitating and Encouraging Teamwork!

- DataJam is a team competition.Teams *boost productivity*, *even out workload*, allow for a *deeper understanding* of the topic, & make the DataJam experience *more fun*!
- Teams should discuss ideas together so everyone is involved.
- Think about splitting up work.
- Progress checks and communication are essential!
- Set up regular meetings with your DataJam Mentor, and communicate between meetings on SLACK. Ask Mentors questions; if they don't know the answer they can seek guidance from The DataJam.

See the *Teamwork Guide* on the *Resources page* of The DataJam website.

Proposals and Initial Team Work

Coaching Strategies: We will discuss how to help students generate strong research questions, hypotheses, and analytical ideas.

1. Pick a topic close to "home"

2. Is this idea too broad?

3. Does this idea have data available?

local interest to your community.

Completed High School DataJam Projects

2016 2017 2018 2019 2020

2021 2022 2023 2024 2025

2025 POSTERS

Avonworth Team 1
The American Dream: Civic Engagement Across the White-Picket Fence

Avonworth Team 2
Dollars and Sense: The SAT-Income Equation

Avonworth Team 3
Moving Around the 'Burgh

Bisbee Team 1
Beaver Data Poster

Bisbee Team 2
Water Scarcity, Environmental Factors, and the Beaver Populations in the Sand Pedro River

Bisbee Team 3
The Migration of the Sand Pedro Beavers and its Effects on the Mexican Side

Canyon Crest Academy Team 6
Erasing the Unwanted: The Link Between Memory Suppression and Psychological Disorders

Canyon Crest Academy Team 7
EVALUATING THE DISPARITY BETWEEN COVID-19 VACCINE ADMINISTRATION RATES IN U.S. COUNTIES WITH LOW VS. HIGH MEDIAN HOUSEHOLD INCOMES

Canyon Crest Academy Team 8
Examining Racial Disparities in Health Care Access Specifically Whether the Individual is Insured or Uninsured Within Cities

Canyon Crest Academy Team 9
Correlation between AI Adoption Level and Job Industry

Canyon Crest Academy Team 10

Keystone Oaks Team 1
The Wizard of Oz(Empic)

Keystone Oaks Team 2
Class Size Beats Cash Prize

Maple Grove Team 1
How Coffee & Energy Drinks Affect YOUR Health

Monarch
Does the Population of Different Neighborhood Effect the Collison Rates in San Diego?

Moon Area High School
AI vs. Reality: Detecting the Markers of AI-Generated Images

Miami Valley Career Technology Center Team 1
Saturn's Moons

Passaic Academy Team 2
How does perceived ease of getting drugs vary from race?

Passaic Academy Team 3
Drug overdoses and unemployment in the New York

Passaic Academy Team 4
Can you be victim to nicotine?

Passaic Academy Team 5
Price of Fast Food vs Median Household Income per County in the US

Pittsburgh Science and Technology Academy
Does Pollution Affect the Organisms Around it?

Preuss Team 1
Is the cost of acquiring an undergraduate degree in the United States growing at a

Proposals and Initial Team Work

Agreeing on a Research Question: This is one of the most difficult steps in a DataJam Project. It requires teamwork, time and investigation.

1. **Suggest that each student on a team writes down 3 ideas for a research question.**
2. **Discuss each idea:**
   a. **Is it a question that can be answered by data analysis?**
   b. **Is it of interest to the whole team?**
   c. **Will the answer be meaningful?**
   d. **Will a dataset be publicly available to answer the question?**

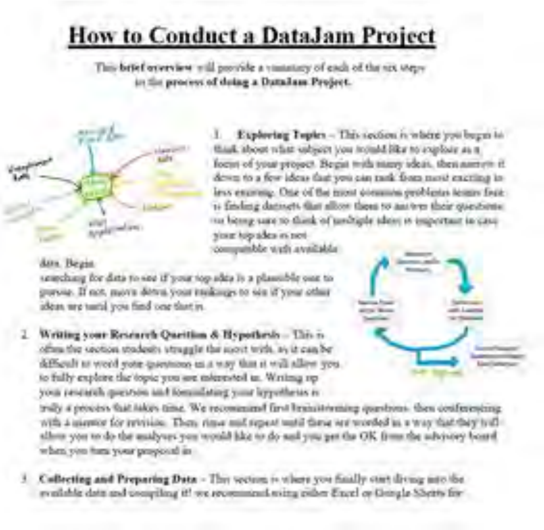
Proposals and Initial Team Work

Finding Data: Guidance on finding and getting data from the web using resources like the DataJam website and data.gov.

1. **Check the DataJam website's Datasets page**
2. **Do a quick google search**
3. **Ask AI like ChatGPT – phrasing is important**

Let's take a tour through the Datasets page.

Resources on The DataJam Website for getting started with a DataJam Proposal!



How to Conduct a DataJam Project

This **brief overview** will provide a summary of each of the six steps in the **process of doing a DataJam Project.**

1. **Exploring Topics** – This section is where you begin to think about what subject you would like to explore as a focus of your project. Begin with many ideas, then narrow it down to a few ideas that you can rank from most exciting to less exciting. One of the most common problems teams face is finding datasets that allow them to answer their questions so being sure to think of multiple ideas is important in case your top idea is not compatible with available data. Begin searching for data to see if your top idea is a plausible one to pursue. If not, move down your rankings to see if your other ideas are until you find one that is.
2. **Writing your Research Question & Hypothesis** – This is often the section students struggle the most with, as it can be difficult to word your questions in a way that it will allow you to fully explore the topic you are interested in. Writing up your research question and formulating your hypothesis is truly a process that takes time. We recommend first brainstorming questions, then conferencing with a mentor for revision. Then rinse and repeat until these are worded in a way that they will allow you to do the analyses you would like to do and you get the OK from the advisory board when you turn your proposal in.
3. **Collecting and Preparing Data** – This section is where you finally start diving into the available data and compiling it! we recommend using either Excel or Google Sheets for

1. **Identify a question**

1. **Find data**

1. **Practice filling out the proposal sheet.**

Proposals and Initial Team Work

Your Task: You will create a brief, outline-level mini-DataJam proposal to practice these skills.

Experience a “Mock DataJam” as a Team

We will now break out into group(s) and practice interacting with a mentor during each stage of The DataJam competition.

THE DATAJAM

Regroup and Final Thoughts

- What was the most challenging part of creating your mini-proposal?

- Based on your experience today, what do you anticipate will be the biggest challenge for your student teams?

- What's one new idea or piece of information you'll take away from this workshop that you're excited to apply with your team?

- Why is it so important to have students define their research question before they start looking for data?